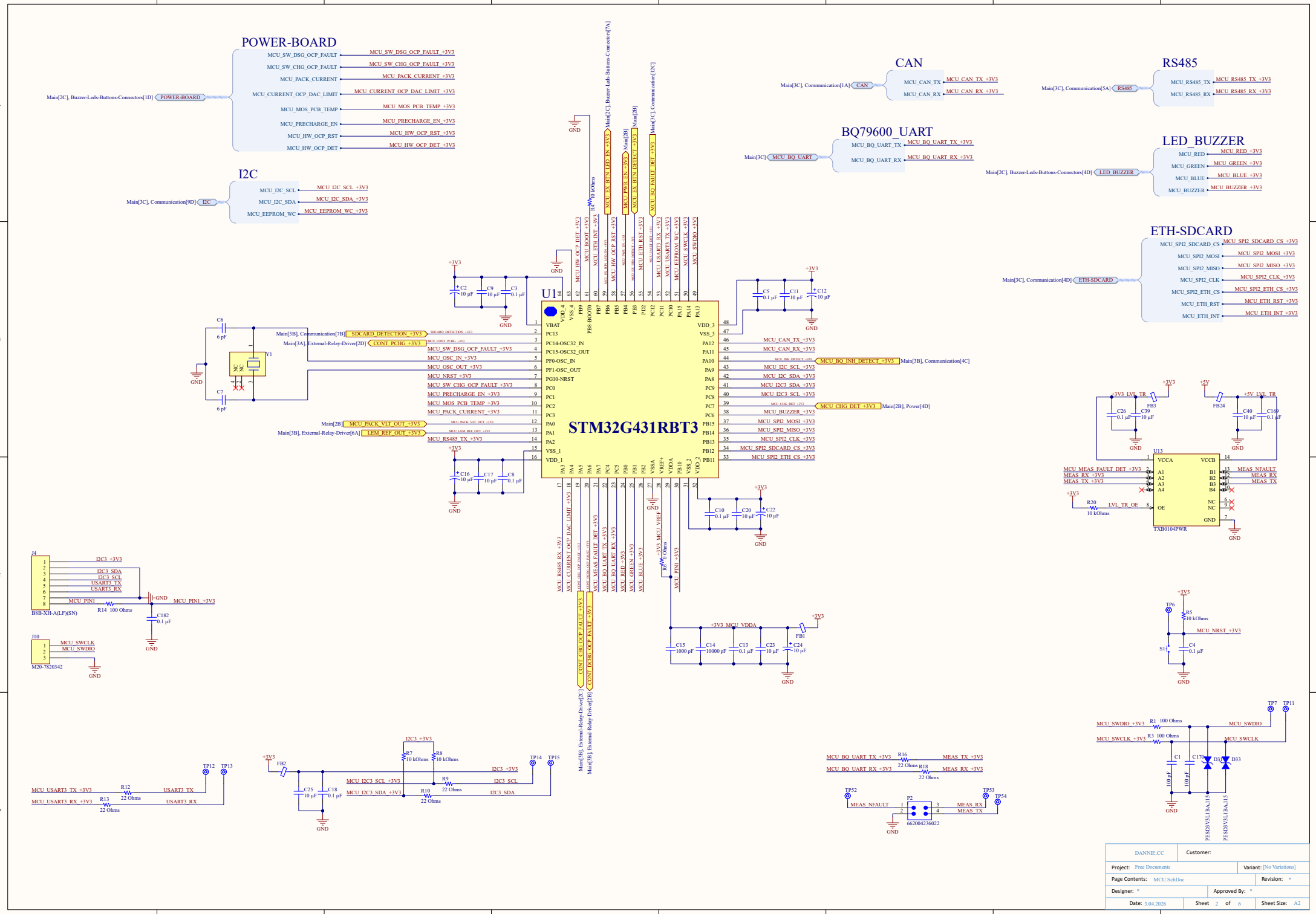
POWER-BOARD
Main[2C], Buzzer-Leds-Buttons-Connectors[1D]
POWER-BOARD
MCU_SW_DSG_OCP_FAULT
MCU_SW_CHG_OCP_FAULT
MCU_PACK_CURRENT
MCU_CURRENT_OCP_DAC_LIMIT
MCU_MOS_PCB_TEMP
MCU_PRECHARGE_EN
MCU_HW_OCP_RST
MCU_HW_OCP_DET
MCU_SW_DSG_OCP_FAULT_+3V3
MCU_SW_CHG_OCP_FAULT_+3V3
MCU_PACK_CURRENT_+3V3
MCU_CURRENT_OCP_DAC_LIMIT_+3V3
MCU_MOS_PCB_TEMP_+3V3
MCU_PRECHARGE_EN_+3V3
MCU_HW_OCP_RST_+3V3
MCU_HW_OCP_DET_+3V3
I2C
Main[3C], Communication[9D]
I2C
MCU_I2C_SCL
MCU_I2C_SDA
MCU_EEPROM_WC
MCU_I2C_SCL_+3V3
MCU_I2C_SDA_+3V3
MCU_EEPROM_WC_+3V3
CAN
Main[3C], Communication[1A]
CAN
MCU_CAN_TX
MCU_CAN_RX
MCU_CAN_TX_+3V3
MCU_CAN_RX_+3V3
RS485
Main[3C], Communication[5A]
RS485
MCU_RS485_TX
MCU_RS485_RX
MCU_RS485_TX_+3V3
MCU_RS485_RX_+3V3
BQ79600_UART
Main[3C]
MCU_BQ_UART
MCU_BQ_UART_TX
MCU_BQ_UART_RX
MCU_BQ_UART_TX_+3V3
MCU_BQ_UART_RX_+3V3
LED_BUZZER
Main[2C], Buzzer-Leds-Buttons-Connectors[4D]
LED_BUZZER
MCU_RED
MCU_GREEN
MCU_BLUE
MCU_BUZZER
MCU_RED_+3V3
MCU_GREEN_+3V3
MCU_BLUE_+3V3
MCU_BUZZER_+3V3
ETH-SDCARD
Main[3C], Communication[4D]
ETH-SDCARD
MCU_SPI2_SDCARD_CS
MCU_SPI2_MOSI
MCU_SPI2_MISO
MCU_SPI2_CLK
MCU_SPI2_ETH_CS
MCU_ETH_RST
MCU_ETH_INT
MCU_SPI2_SDCARD_CS_+3V3
MCU_SPI2_MOSI_+3V3
MCU_SPI2_MISO_+3V3
MCU_SPI2_CLK_+3V3
MCU_SPI2_ETH_CS_+3V3
MCU_ETH_RST_+3V3
MCU_ETH_INT_+3V3
U1
STM32G431RBT3
VBAT
PC13
PC14-OSC32_IN
PC15-OSC32_OUT
PF0-OSC_IN
PF1-OSC_OUT
PG10-NRST
PC0
PC1
PC2
PC3
PA0
PA1
PA2
VSS_1
VDD_1
VDD_4
VSS_4
PB9
PB8-BOOT0
PB7
PB6
PB5
PB4
PB3
PD2
PC12
PC11
PC10
PA15
PA14
PA13
VDD_3
VSS_3
PA12
PA11
PA10
PA9
PA8
PC9
PC8
PC7
PC6
PB15
PB14
PB13
PB12
PB11
PA3
PA4
PA5
PA6
PA7
PC4
PC5
PB0
PB1
PB2
VSSA
VREF+
VDDA
PB10
VSS_2
VDD_2
MCU_HW_OCP_DET_+3V3
MCU_BOOT_+3V3
MCU_ETH_INT_+3V3
MCU_HW_OCP_RST_+3V3
MCU_ETH_RST_+3V3
MCU_USART3_RX_+3V3
MCU_USART3_TX_+3V3
MCU_EEPROM_WC_+3V3
MCU_SWCLK_+3V3
MCU_SWDIO_+3V3
MCU_CAN_TX_+3V3
MCU_CAN_RX_+3V3
MCU_I2C_SCL_+3V3
MCU_I2C_SDA_+3V3
MCU_I2C3_SDA_+3V3
MCU_I2C3_SCL_+3V3
MCU_BUZZER_+3V3
MCU_SPI2_MOSI_+3V3
MCU_SPI2_MISO_+3V3
MCU_SPI2_CLK_+3V3
MCU_SPI2_SDCARD_CS_+3V3
MCU_SPI2_ETH_CS_+3V3
MCU_RS485_RX_+3V3
MCU_CURRENT_OCP_DAC_LIMIT_+3V3
MCU_MEAS_FAULT_DET_+3V3
MCU_BQ_UART_TX_+3V3
MCU_BQ_UART_RX_+3V3
MCU_RED_+3V3
MCU_GREEN_+3V3
MCU_BLUE_+3V3
MCU_PIN1_+3V3
MCU_BQ_INH_DETECT_+3V3
MCU_CHG_DET_+3V3
MCU_PACK_VLT_OUT_+3V3
LEM_REF_OUT_+3V3
SDCARD_DETECTION_+3V3
CONT_PCHG_+3V3
MCU_PACK_VLT_OUT_+3V3
MCU_LEM_REF_OUT_+3V3
MCU_CHG_DET_+3V3
MCU_ETH_INT_+3V3
MCU_ETH_RST_+3V3
MCU_BQ_FAULT_DET_+3V3
MCU_PWR_EN_+3V3
MCU_LED_BTN_DETECT_+3V3
MCU_LED_BTN_LED_EN_+3V3
CONT_CHG_OCP_FAULT_+3V3
CONT_DCHG_OCP_FAULT_+3V3
Main[2C], Buzzer-Leds-Buttons-Connectors[7A]
Main[2B]
Main[3C], Communication[12C]
Main[3B], Communication[7B]
Main[3A], External-Relay-Driver[2D]
Main[2B]
Main[3B], External-Relay-Driver[6A]
Main[3B], Communication[4C]
Main[2B], Power[4D]
Main[3C], External-Relay-Driver[3C]
Main[3B], External-Relay-Driver[2B]
R4 0 kOhms
R20 0 Ohms
C2 10 µF
C9 10 µF
C3 0.1 µF
C6 6 pF
C7 6 pF
Y1
C16 10 µF
C17 10 µF
C8 0.1 µF
C5 0.1 µF
C11 10 µF
C12 10 µF
C10 0.1 µF
C20 10 µF
C22 10 µF
C15 1000 pF
C14 10000 pF
C13 0.1 µF
C23 10 µF
C24 10 µF
FB1
J4
B8B-XH-A(LF)(SN)
I2C3_+3V3
I2C3_SDA
I2C3_SCL
USART3_TX
USART3_RX
MCU_PIN1
MCU_PIN1_+3V3
R14 100 Ohms
C182 0.1 µF
J10
M20-7820342
MCU_SWCLK
MCU_SWDIO
TP12
TP13
TP14
TP15
FB2
C25 10 µF
C18 0.1 µF
R7 10 kOhms
R8 10 kOhms
R9 22 Ohms
R10 22 Ohms
R12 22 Ohms
R13 22 Ohms
MCU_USART3_TX_+3V3
MCU_USART3_RX_+3V3
USART3_TX
USART3_RX
MCU_I2C3_SCL_+3V3
MCU_I2C3_SDA_+3V3
I2C3_SCL
I2C3_SDA
U13
TXB0104PWR
VCCA
VCCB
A1
A2
A3
A4
B1
B2
B3
B4
OE
NC
GND
FB3
FB24
C26 0.1 µF
C39 10 µF
C40 10 µF
C169 0.1 µF
R20 10 kOhms
LVL_TR_OE
MCU_MEAS_FAULT_DET_+3V3
MEAS_RX_+3V3
MEAS_TX_+3V3
MEAS_NFAULT
MEAS_RX
MEAS_TX
TP6
R5 10 kOhms
S14
C4 0.1 µF
MCU_NRST_+3V3
TP7
TP11
R1 100 Ohms
R3 100 Ohms
C1 100 pF
C170 100 pF
D31
D33
PESD3V3L1BA,115
MCU_BQ_UART_TX_+3V3
MCU_BQ_UART_RX_+3V3
R16 22 Ohms
R18 22 Ohms
TP52
TP53
TP54
P2
662004236022
MEAS_NFAULT
MEAS_RX
MEAS_TX
DANNIE.CC
Customer:
Project: Free Documents
Variant: [No Variations]
Page Contents: MCU.SchDoc
Revision: *
Designer: *
Approved By: *
Date: 3.04.2026
Sheet 2 of 6
Sheet Size: A2